

Protein synthesis Inhibitors

PHAM201

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Objectives

- Differentiate medications that are protein synthesis inhibitors and their mechanisms of action.
- Formulate appropriate protein synthesis inhibitors used for different infections with explanation of time-dependent and concentration-dependent killing actions of antibiotics and what's meant by post antibiotic effect.
- Describe the pharmacokinetics and adverse effects of protein synthesis inhibitors

CELL MEMBRANE

CELL WALL

Inhibitors of cell membrane function

- Amphotericin B
- Isoniazid
- Polymyxins

Inhibitors of nucleic acid function or synthesis

- Fluoroquinolones
- Rifampin

Inhibitors of protein synthesis

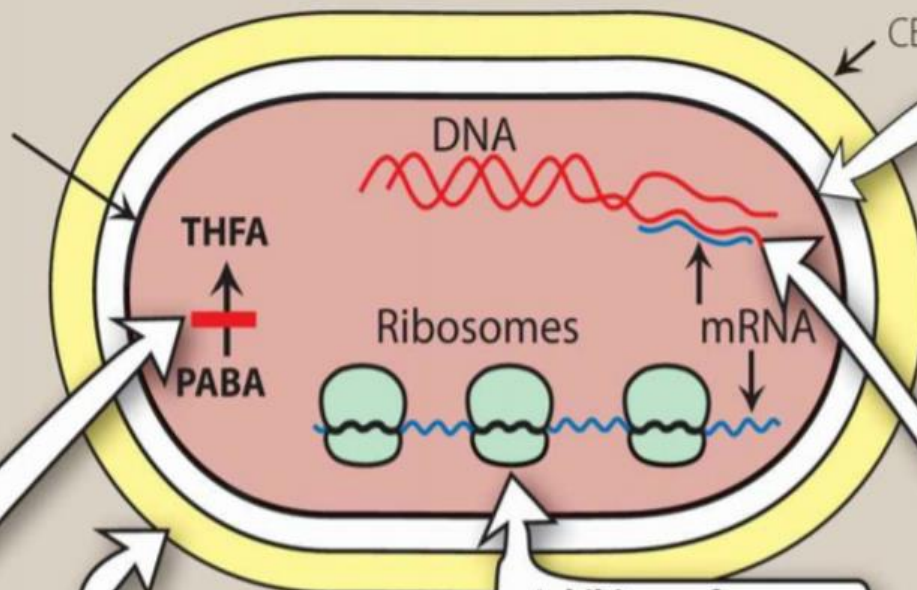
- Aminoglycosides
- Chloramphenicol
- Clindamycin
- Macrolides
- Oxazolidinones
- Tetracyclines

Inhibitors of cell wall synthesis

- β -Lactams
- Daptomycin
- Fosfomycin
- Lipoglycopeptides
- Vancomycin

Inhibitors of metabolism

- Sulfonamides
- Trimethoprim



Protein synthesis inhibitors

TETRACYCLINES

Demeclocycline DECLOMYCIN

Doxycycline VIBRAMYCIN

Minocycline MINOCIN

Tetracycline

GLYCYLCYCLINES

Tigecycline TYGACIL

AMINOGLYCOSIDES

Amikacin

Gentamicin GARAMYCIN

Neomycin NEO-FRADIN

Streptomycin

Tobramycin TOBREX

MACROLIDES/KETOLIDES

Azithromycin ZITHROMAX

Clarithromycin BIAXIN

Erythromycin VARIOUS

Telithromycin KETEK

MACROCYCLIC

Fidaxomicin DIFICID

LINCOSAMIDES

Clindamycin CLEOCIN

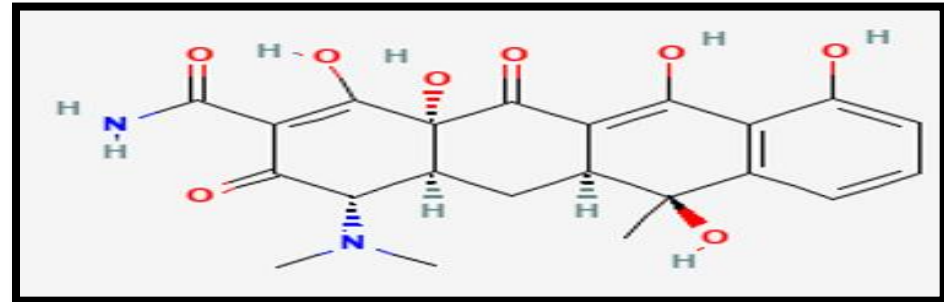
OXAZOLIDINONES

Linezolid ZYVOX

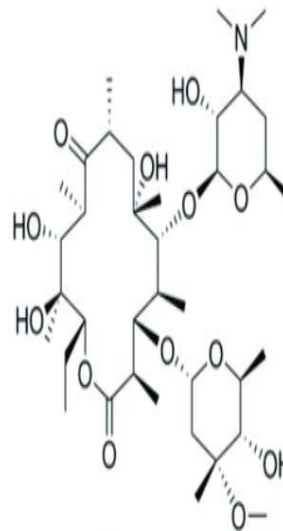
OTHERS

Chloramphenicol CHLOROMYCETIN

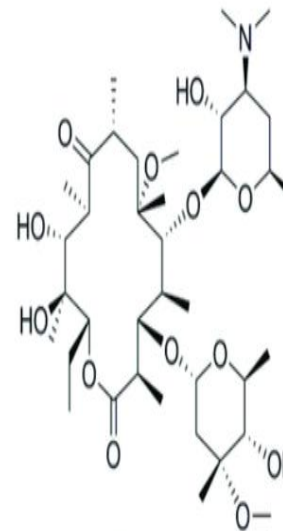
Quinupristin/Dalfopristin SYNERCID



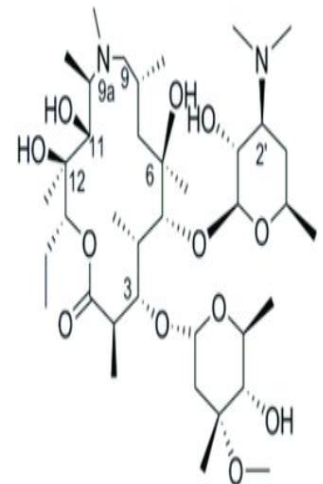
Chemical structure of tetracyclines



Erythromycin



Clarithromycin



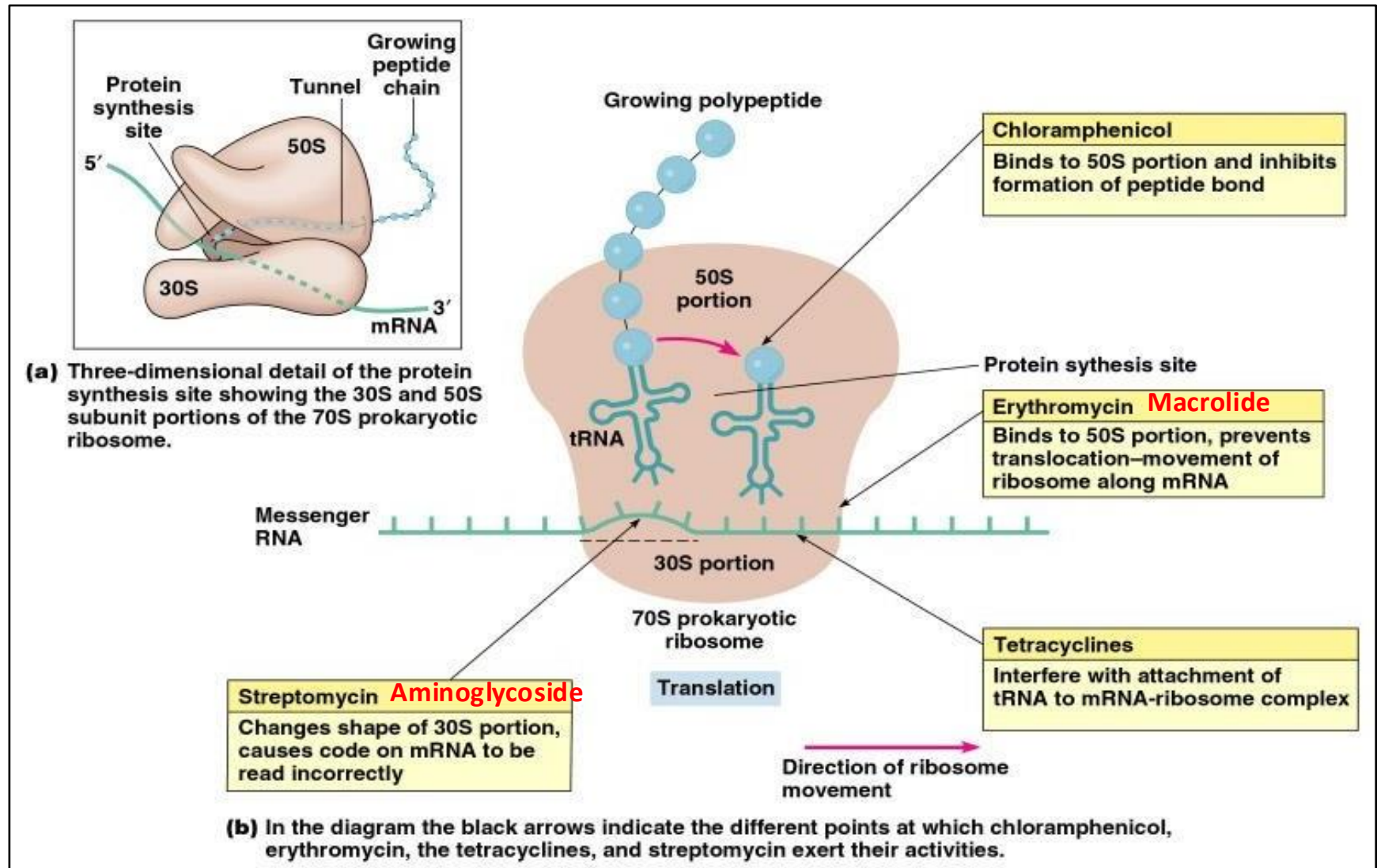
Azithromycin

Chemical structure of macrolide antibiotics.

Figure 39.1

Summary of protein synthesis inhibitors.

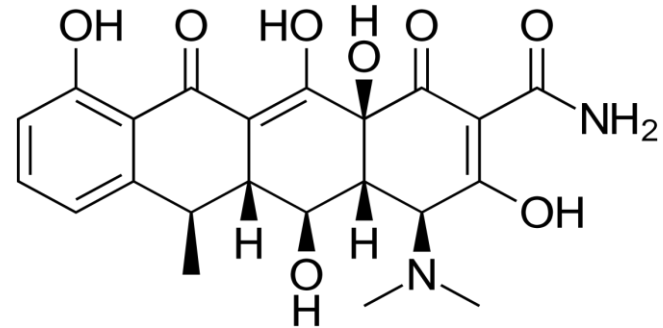
Inhibitors of Protein Synthesis: Mechanisms



Tetracyclines

Members:

- Tetracycline
- Doxycycline
- Minocycline
- Demeclocycline



Mechanism of action: Reversibly bind to 30 S ribosomal bacterial subunits leading to inhibition of binding of tRNA to the mRNA–ribosome complex, thereby inhibiting bacterial protein synthesis

They are *bacteriostatic*.

Tetracyclines uses

PEPTIC ULCER DISEASE

- Helicobacter pylori is a common cause of peptic ulcer disease.
- Treatment with a combination of *bismuth*, *metronidazole*, *tetracycline*, and a proton pump inhibitor is a highly effective regimen for eradication of H. pylori

LYME DISEASE

- This is a spirochetal infection caused by Borrelia burgdorferi. The disease is transmitted by the bite of infected ticks.
- Infection results in skin lesions, headache, and fever, followed by meningoencephalitis and, eventually, arthritis.
- A bull's-eye pattern rash with a red outer ring, called erythema migrans is a hallmark of Lyme disease
- *Doxycycline* is one of the preferred therapeutic options.

MYCOPLASMA PNEUMONIAE

- Mycoplasma pneumoniae, or walking pneumonia, is a common cause of community-acquired pneumonia in young adults and in people who live in close confines, such as in military camps.
- Treatment with a macrolide or *doxycycline* is effective.

Gram (+) cocci

Staphylococcus aureus
(including methicillin-resistant strains)
Streptococcus pneumoniae

Gram (+) bacilli

Bacillus anthracis

Gram (-) cocci

Gram (-) rods

Brucella species*
Helicobacter pylori
Vibrio cholerae
Yersinia pestis

Anaerobic organisms

Clostridium perfringens
Clostridium tetani

Spirochetes

Borrelia burgdorferi
Leptospira interrogans
Treponema pallidum

Mycoplasma

Mycoplasma pneumoniae

Chlamydia

Chlamydia species

Other

Rickettsia rickettsii

CHOLERA

- Cholera is caused by Vibrio cholerae ingested in fecally contaminated food or water.
- The organism multiplies in the gastrointestinal tract, where it secretes an enterotoxin that produces diarrhea.
- Treatment includes *doxycycline*, which reduces the number of intestinal vibrios, and fluid replacement.

CHLAMYDIAL INFECTIONS

- Chlamydia trachomatis is a major cause of sexually transmitted disease in the United States. It causes nongonococcal urethritis, pelvic inflammatory disease, and lymphogranuloma venereum.
- Chlamydia psittaci causes psittacosis, which usually takes the form of pneumonia. Other clinical forms include hepatitis, myocarditis, and coma.
- *Doxycycline* or *azithromycin* is used to treat chlamydial infections.

ROCKY MOUNTAIN SPOTTED FEVER

- This disease, caused by Rickettsia rickettsii, is characterized by fever, chills, and aches in bones and joints.
- Response to tetracyclines is prompt if the drug is started early in the disease process.

Tetracyclines uses



- **Tetracyclines** affect a wide variety of microbial species and are referred to as broad-spectrum antibiotics
- **Broad-spectrum**: effective against gram-positive and gram-negative bacteria, protozoa, spirochetes, mycobacteria, and atypical species.

Examples: Acne, Chlamydia infections, cholera (**doxycycline**), mycoplasma pneumonia, Lyme disease, H pylori, Rocky Mountain spotted fever (**Rickettsia**) and anaerobic organisms e.g. clostridia.

Demeclocycline: is used in treatment of syndrome of *inappropriate secretion of ADH* (**SIADH**) (excess production of ADH producing)

Tetracyclines

Pharmacokinetics:

Absorption:

- Incompletely absorbed when orally administered except for **minocycline** and **doxycycline** which are almost completely absorbed.
- Absorption is decreased by **food** due to formation of non absorbable chelates of tetracyclines with Ca^{++} , Mg^{++} , Al^{+++} and iron.

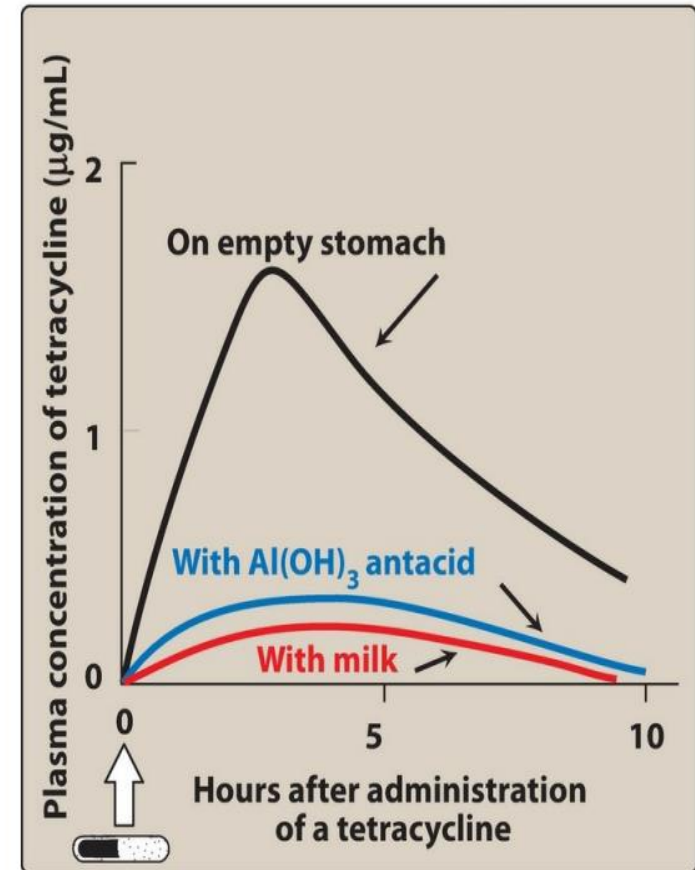


Figure 30.5 Effect of antacids and milk on the absorption of tetracyclines.

Tetracyclines

Pharmacokinetics:

Distribution:

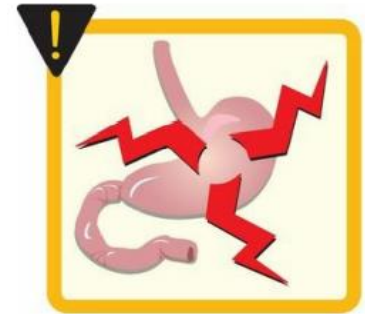
- They penetrate BBB in *insufficient concentrations* except **minocycline** which is effective **only for meningococcal carrier states** *not* for CNS infections.
- Bind to tissues undergoing calcification (**teeth** and **bones**) or to tumors that have **a high calcium content**.
- *So they concentrated in bone and teeth* → deformities in children (up to 8 years).
- Cross the placenta and affect fetal bones and teeth.

Metabolism and excretion:

They are eliminated unchanged in the urine (via kidney) **except doxycycline**, which is preferred in patients with renal dysfunction, as it is primarily eliminated via the bile into the feces.

Adverse effects and contraindications:

Gastric discomfort due to gastric irritation (non compliance)



GI disturbance

Teeth discoloration and hypoplasia and a temporary stunting of growth (**bone hypoplasia**).

- **Tetracyclines bind to calcium in growing bones and teeth** → can **discolor teeth**. So, **should be avoided during pregnancy (inhibit fetal bone growth) and in children < 8 years old**.
- They must not be taken with any drug or food containing Ca (Cheese & milk).



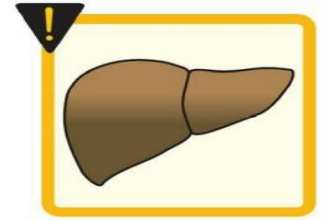
Deposition of drug in bones and teeth



Avoid in pregnancy

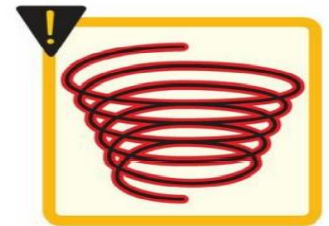
Adverse effects and contraindications:

Fatal Hepatotoxicity (in renal dysfunction or pregnant females receiving high doses).



Liver failure

Vestibular dysfunction: Dizziness, vertigo, and tinnitus may occur particularly with **minocycline**, which concentrates in the endolymph of the ear and affects function.



Vertigo

Phototoxicity



Phototoxicity

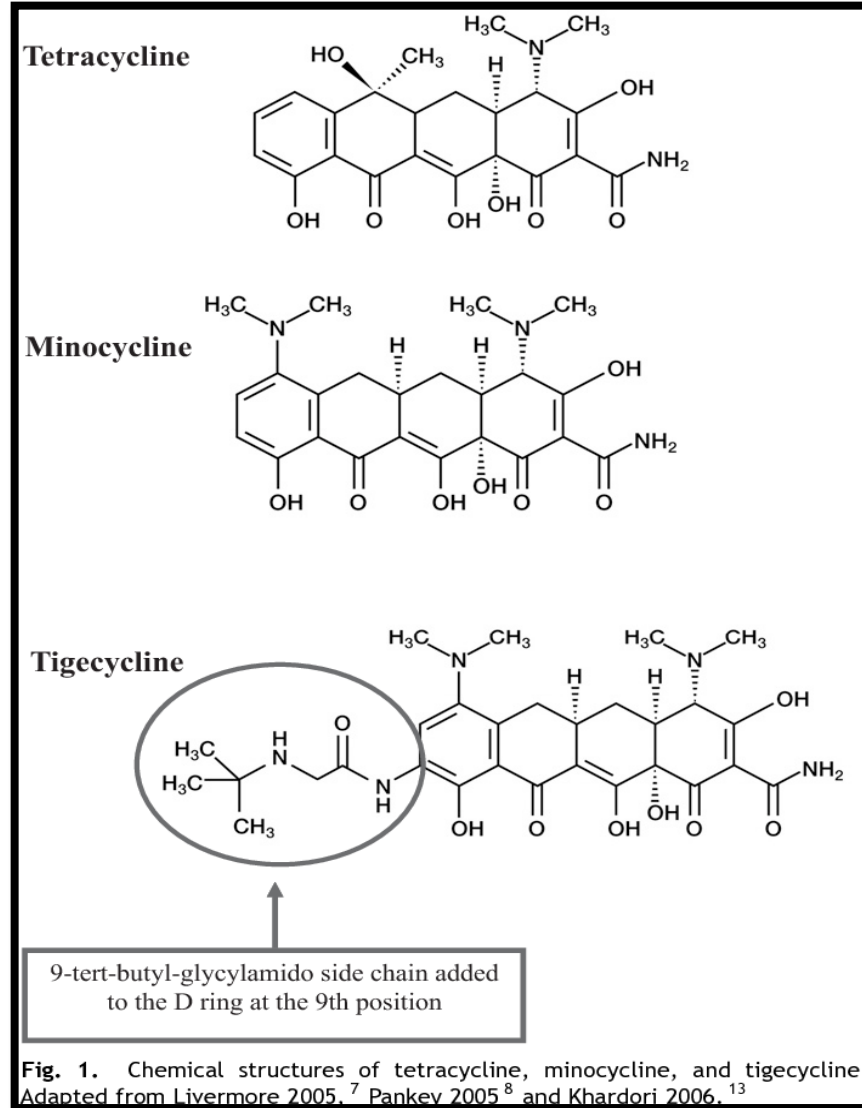
Superinfection

Renal dysfunction, thus all tetracyclines are contraindicated in renal impairment except **doxycycline**

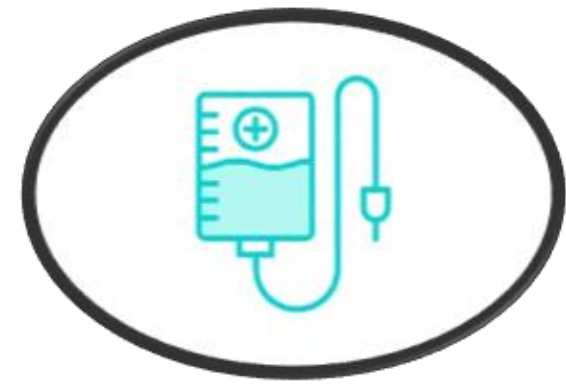
ADDITIONAL BROAD-SPECTRUM DRUGS

Glycylcycline (Tigecycline)

- Derivative of minocycline
- related to tetracyclines



Tigecycline



- It is given by injection (IV)

Mechanism of action: binds to the **30S ribosomal** subunit inhibiting protein synthesis.

Spectrum: Tigecycline has **broad-spectrum activity** against **Gram-positive bacteria**, including **Staphylococci (MRSA)** and **Enterococci (VRE)**, **Gram-negative bacteria**, anaerobes and atypical organisms.

Among gram-negative Bactria, it has **no activity** against the "3 P's": **Pseudomonas, Proteus, Providencia** species.

Tigecycline pharmacokinetics

-It is **lipophilic** and due to the **high** volume of distribution (V_d) of tigecycline, it achieves poor serum concentrations. For this reason → tigecycline **should not** be used for endovascular infections, such as bacteremia or endocarditis.

Clinical Uses of tigecycline

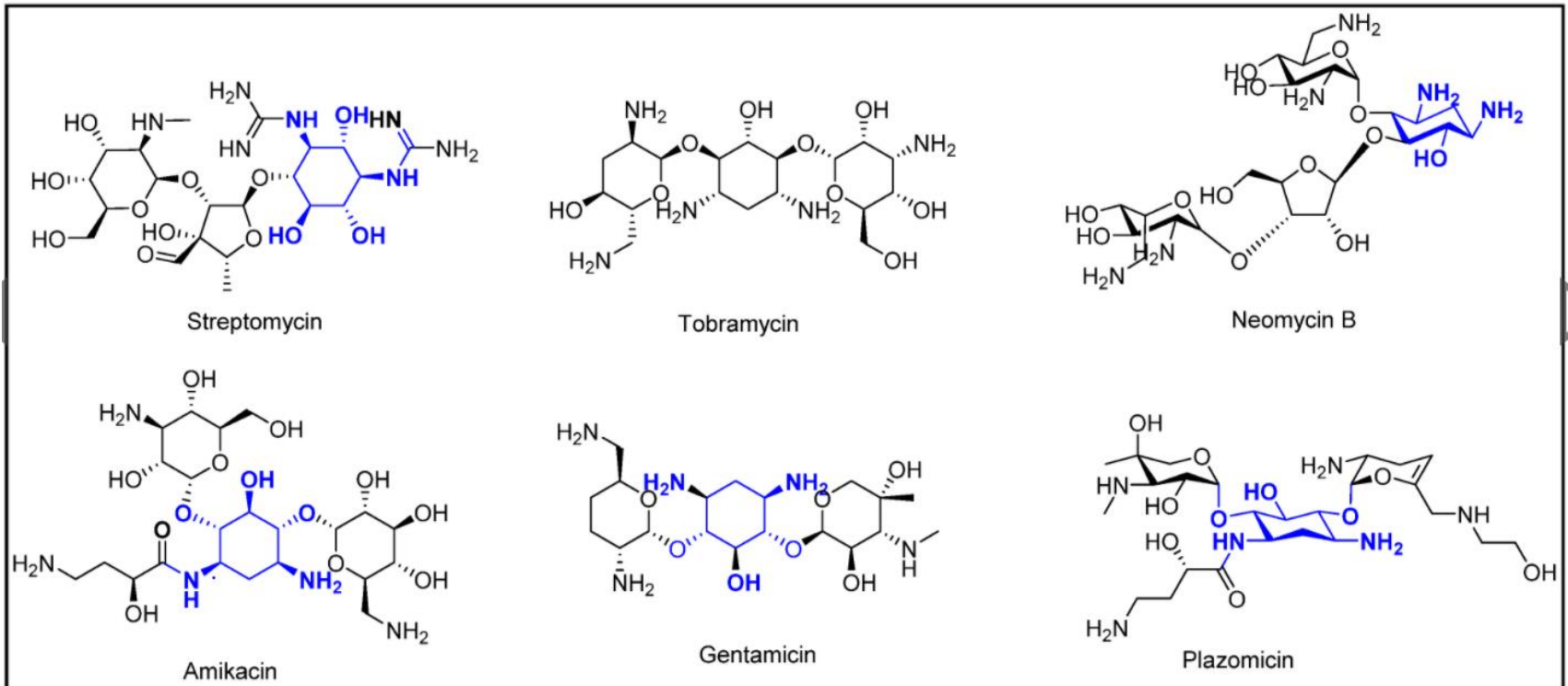
1. **Complicated skin infections**; including abscesses and infected burns or ulcers caused bacteria including *E.coli*, *Enterococcus faecalis*, *MRSA*.
2. **Complicated intra-abdominal infections**, including abscesses, appendicitis, cholecystitis, ..
3. **Community-acquired bacterial pneumonia** caused by *S. pneumoniae*, *H. influenzae* and *Legionella pneumophila*

Tigecycline safety/ side effects & monitoring

- **Side effects like tetracycline:** hepatotoxicity, pancreatitis, photosensitivity, teeth discoloration in children < 8 years old.
- **BOXED WARNING:** Increase risk of death, so use only when alternative treatments are not suitable
- **Drug interaction:**
Tigecycline can increase the INR in patients taking **warfarin**.

Aminoglycosides

The term “aminoglycoside” comes from their structure—two amino sugars joined by a glycosidic linkage to a central hexose nucleus.



Aminoglycosides

Members

- **Streptomycin**
- **Gentamicin**
- **Tobramycin**
- **Amikacin**
- **Neomycin**

Aminoglycosides

Mechanism of action

- **Irreversibly** bind with **30S** ribosomal bacterial subunits leading to inhibition of protein synthesis by:
 - Inhibition of formation of initiation complex.
 - Misreading of mRNA (the genetic code) leading to formation of abnormal proteins.
 - Breaking down of polysomes into non-functional monosomes.

Aminoglycosides

- Aminoglycosides have **concentration-dependent killing**, i.e., increasing concentrations from 4-64 folds the MIC (lowest concentration that inhibits bacterial growth) of the antibiotic cause **increasing proportion of bacteria to be killed and at a more rapid rate.**
- Aminoglycosides have **post antibiotic effect** (persistent inhibition of bacterial growth after concentration of antimicrobial have fallen below MIC) which enables **once daily dosing** (less toxicities and less expense).

The time-dependent antibiotics

Higher concentrations of antibiotics--> do not result in greater kill of organisms. Furthermore, they tend to have minimal to no post antibiotic effect (PAE) e.g. **penicillins, cephalosporins, macrolides and clindamycin, and linezolid.**

Aminoglycosides

Pharmacokinetics

Absorption:

- **Not absorbed orally**
- All aminoglycosides (except **neomycin**) must be given **parenterally** to achieve adequate serum levels because of the highly polar, polycationic structure of the aminoglycosides which prevents adequate absorption after oral administration.
- **Neomycin** is not given parenterally due to **severe nephrotoxicity**. It is administered **topically** for skin infections or **orally** for bowel preparation prior to colorectal surgery.

Aminoglycosides

Distribution:

- They **don't cross BBB** in sufficient concentrations even when meninges are inflamed, but **cross the placental barrier**.
- Concentrated in renal cortex ,and in perilymph & endolymph of inner ear → **nephrotoxicity and ototoxicity**.
- The immediate high plasma concentrations that result from **single daily dosing** are advantageous, e.g. for *acutely ill septicemic* patients.
- **All cross placental barrier and concentrated in fetal plasma.**

Excretion:

Excreted unchanged through **the kidney**, so dose adjustment must be taken when given to patients with renal dysfunction.

Aminoglycosides

Antibacterial spectrum

- Aminoglycosides are unique in that they are bactericidal.
- Effective for most **aerobic gram-negative organisms** (e.g. pseudomonas, E coli, Klebsiella pneumoniae).
- **Ineffective** against **anaerobes**
- Not used for meningitis (they can't cross BBB)
- Used for **empirical** treatment of infections (**synergistic action**) with β -lactam or vancomycin e.g. **infective endocarditis**.
- **Other uses:** **T.B** (Streptomycin, IM), **Plague** (Streptomycin, IM) and **Tularemia** (Gentamicin is effective in treating this disease).

Aminoglycosides

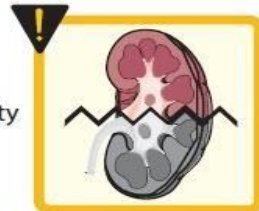
Most important side effects

- **Ototoxicity:** directly related to high peak plasma levels and the duration of treatment. The antibiotic accumulates in the inner ear. Deafness may be **irreversible** and affect developing fetuses.
- **Nephrotoxicity** (*Reversible*).
- **Neuromuscular paralysis:** This adverse effect is associated with a rapid increase in concentrations or concurrent administration with neuromuscular blockers. Patients with myasthenia gravis are particularly at risk. **It is usually reversible by treatment with calcium gluconate or neostigmine** which can reverse the block that causes neuromuscular paralysis.
- **Allergic reactions:** Contact dermatitis is a common reaction to topically applied **neomycin**.

Ototoxicity



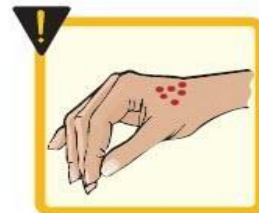
Nephrotoxicity



Paralysis



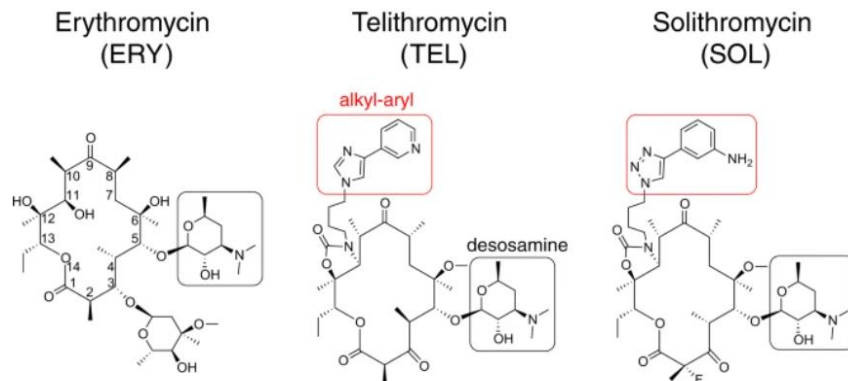
Skin rash



Macrolides

Members:

- **Erythromycin**
- **Clarithromycin** (a methylated form of erythromycin)
- **Azithromycin** (a larger lactone ring)
- **Telithromycin** (a semisynthetic derivative of erythromycin, is a “ketolide” antimicrobial agent)



Macrolides

Mechanism of action

- Bind irreversibly to a site on the 50S subunit of the bacterial ribosome, thus inhibiting translocation steps of protein synthesis.
- **Bacteriostatic**, they may be **bactericidal at higher doses**.



Macrolides

Antibacterial spectrum & uses:

CORYNEBACTERIUM DIPHTHERIAE

- *Erythromycin* or *penicillin* is used to eliminate the carrier state.

Gram (+) cocci

Streptococcus pyogenes
Streptococcus pneumoniae

Gram (+) bacilli

Corynebacterium diphtheriae

Gram (-) cocci

Moraxella catarrhalis
Neisseria gonorrhoeae

Gram (-) rods

Bordetella pertussis
Campylobacter jejuni
Haemophilus influenzae
Legionella pneumophila

Anaerobic organisms

Spirochetes

Treponema pallidum

Mycoplasma

Mycoplasma pneumoniae
Ureaplasma urealyticum

Chlamydia

Chlamydia pneumoniae
Chlamydia psittaci
Chlamydia trachomatis

Other

Mycobacterium avium complex

LEGIONNAIRES DISEASE (LEGIONELLOSIS)

- Undiagnosed and asymptomatic infections are common.
- Fluoroquinolones or *azithromycin* are preferred therapeutic options.

MYCOPLASMA PNEUMONIA

- Called "atypical" pneumonia because causative mycoplasma escape isolation by standard bacteriologic techniques.
- *Azithromycin* or *doxycycline* are preferred therapeutic options.

CHLAMYDIAL INFECTIONS

- *Azithromycin* or *doxycycline* are preferred therapeutic options.

MYCOBACTERIUM AVIUM COMPLEX

- *Clarithromycin* in combination with *rifampin* and *ethambutol* is preferred treatment of MAC infections. *Azithromycin* is an alternative to *clarithromycin* in this regimen.
- Once-weekly *azithromycin* is used as MAC prophylaxis in patients with AIDS.

Macrolides

Antibacterial spectrum & uses:

Erythromycin:

- Drug of choice for **patients allergic to penicillins**
- **Gm +ve cocci and bacilli** as an alternative to penicillins/cephalosporins and tetracyclines.
- **Chlamydia** (of choice in pregnancy), **mycoplasma** (in children), and spirochetes.

Macrolides

Antibacterial spectrum & uses:

Clarithromycin:

- Clarithromycin has activity similar to erythromycin, but it is also effective against *Haemophilus influenzae*.
- Its activity against intracellular pathogens, such as Chlamydia, Legionella, Moraxella, and *Helicobacter pylori* is higher than that of *erythromycin*.

Azithromycin: (it has very long duration, taken once daily)

- It is more active against respiratory infections due to *Haemophilus influenzae*.
- Preferred in *urogenital Chlamydia* infection (urethritis).
- Drug of choice in *Legionellosis*.

Macrolides

Pharmacokinetics

Absorption:

All are absorbed after **oral** administration , **except erythromycin** (enteric coated or **esterified forms** are available).

Distribution:

- Erythromycin diffuses in all body fluids except the CSF. It accumulates in macrophages & neutrophil.
- **Azithromycin** has the **longest half life**.

Metabolism: via Liver.

Excretion:

Mainly hepatic (NOT excreted in kidney)

	Erythro- mycin	Clarithro- mycin	Azithro- mycin	Telithro- mycin
Oral absorption	Yes	Yes	Yes	Yes
Half-life (hours)	2	3.5	68	10
Conversion to an active metabolite	No	Yes	No	Yes
Percent excretion in urine	< 15	30–50	< 10	13

Macrolides

Side effects

- **Cholestatic jaundice:** especially with erythromycin estolate. (**All should be avoided or used cautiously in liver dysfunction**).
- **Ototoxicity:** Transient deafness with *erythromycin*, especially at high dosages. *Azithromycin* has also been associated with irreversible hearing loss.
- **QT interval prolongation**
- **Inhibition of P450 in the liver: leading to drug interaction**



Ototoxicity



QTc prolongation

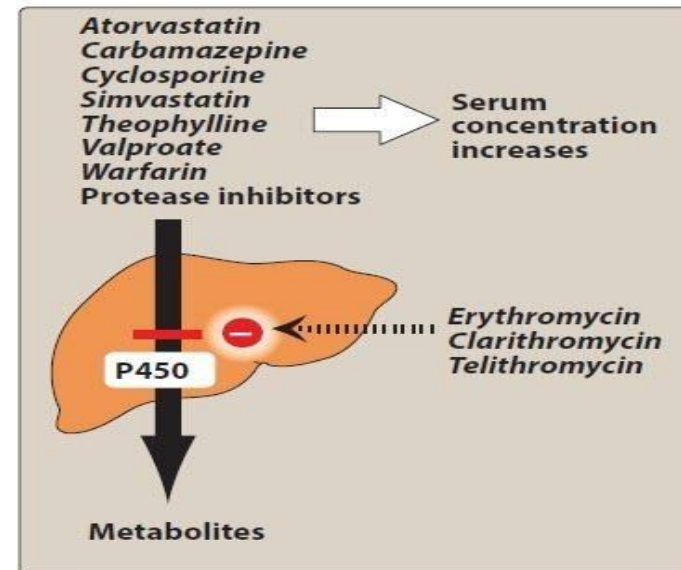


Figure 39.14

Inhibition of the cytochrome P450 system by *erythromycin*,³² *clarithromycin*, and *telithromycin*.

Clindamycin

Similar to macrolides (bacteriostatic with the same mechanism).

Uses:

1. It has activity against most Gram-positive bacteria
2. Anaerobes (e.g. *Bacteroids fragilis* in *penetrating wounds*).
3. Used in **bone infections** (good penetration into bone).

***C. difficile* Infection**

Vancomycin (oral)

Fidaxomicin

Metronidazole

Clostridium difficile infection

- Antibiotics kill normal, healthy GI flora along with the pathogens they are targeting. This results in overgrowth of drug-resistant organisms and can lead to superinfections, such as **Clostridioides** (previously **Clostridium difficile infection “CDI”**).
- Inactive C. difficile spores are present in normal GI flora. When an antibiotic kills off the normal flora, C. difficile spores can become activated, producing toxins **that inflame the GI mucosa**.

Clostridium difficile infection

- Symptoms can be **mild (loose stools and abdominal cramping)** to **severe (pseudomembranous colitis that can require colectomy and can be fatal)**.
- In recent years, C. difficile infections have become more common, more severe and more difficult to treat!!
- All antibiotics have a warning for the risk of CDI, but the risk is highest with broad-spectrum penicillins and cephalosporins, quinolones, carbapenems, and **clindamycin**, which has **a boxed warning**.

Chloramphenicol

Mechanism of action

- Binds reversibly to the bacterial **50S ribosomal** subunit and inhibits protein synthesis at the peptidyl transferase reaction.

Most serious side effects:

- ❑ **Anemias:** Patients may experience dose- related anemia, hemolytic anemia (seen in **patients with glucose-6-phosphate dehydrogenase deficiency**), and aplastic anemia.
- ❑ **Gray baby syndrome:**
 - Neonates have a decreased ability to excrete the drug, which accumulates to levels that leads to bone marrow suppression and cyanosis (hence the term “**gray baby**”).
 - **The reason:** Chloramphenicol is metabolized and inactivated via conjugation in the liver by glucuronyl transferase. The glucuronyl conjugate is easily excreted by the kidneys. Neonates and infants have decreased glucuronyl transferase activity, allowing chloramphenicol to build up to toxic levels.

Linezolid

Mechanism of action

Linezolid binds to the bacterial **23S ribosomal RNA** of the 50S subunit, thereby inhibiting the formation of the 70S initiation complex.

Adverse effects

- Possess a nonselective monoamine oxidase activity (linezolid **inhibits MAO**) and may lead to **serotonin syndrome** if given concomitantly with large quantities *of tyramine-containing foods*, selective serotonin reuptake inhibitors, or monoamine oxidase inhibitors.
- Irreversible peripheral neuropathies and optic neuritis (causing blindness) with >28 days of use.

Others: Metronidazole

- A mitroimidazole, trade name (**Flagyl**).
- MOA: breaks the bacterial or parasitic DNA.
- Spectrum: Gram – anaerobes.
- Important side effect: **metallic taste**

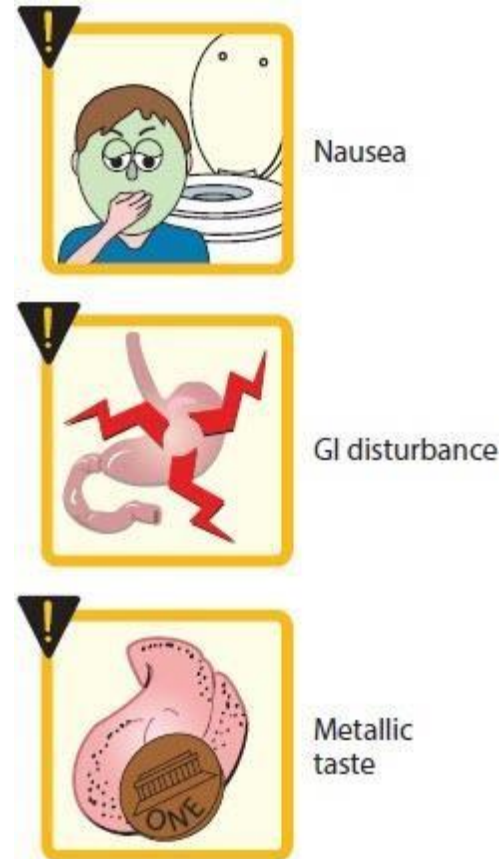


Figure 43.3
Adverse effects of
metronidazole.



THANK YOU!